BIBEK BHANDARI

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Research Interests

Large Language Models (LLMs) · Generative AI · LLM Agents · Trustworthy AI · Natural Language Processing · Human–AI Collaboration · Deep Learning · Transformer Architectures

Education

CMR Institute of Technology, Bengaluru 2022 – 2026
B.E. in Computer Science and Engineering *CGPA:* ***8.66/10***
Machine Learning · Artificial Intelligence · Data Structures and Applications · Design and Analysis of Algorithms · Parallel Computing · Operating Systems · Compiler Design

Research Experience & Publications

Hallucination Detection and Intervention in LLM Agents *Independent Research | Sept 2026 – Present*

- Investigating the robustness of hallucination detection signals in multi-step LLM agents under **adversarial distribution shifts**, focusing on how detection reliability changes as errors propagate across reasoning steps.
- Developing detection-guided intervention mechanisms to identify and mitigate hallucinations before they propagate through subsequent agentic steps, with evaluation across multiple models, tasks, and distribution shifts.

Recursive Fine-Tuning and Model Degradation in Biomedical QA Systems *Preprint | arXiv:2609.34257 | 2026*

- Investigated model degradation from recursive fine-tuning on synthetic biomedical QA data using **LLMs, QLoRA, Deep Learning, and Hugging Face Transformers** across multiple training generations and random seeds.
- Evaluated lexical, semantic, and biomedical entity-level changes using **PubMedQA, BERT-based semantic metrics, biomedical NER, and statistical analysis**.
- Analyzed changes in model diversity, domain knowledge retention, and answer quality across recursive generations to characterize the effects of synthetic data accumulation.
- **arXiv:2609.34257**

Projects

Compute-Efficient LoRA Objective Selection *Qwen3-0.6B, LoRA, MLX*

- Developed and evaluated a shared-prefix LoRA training strategy for comparing token-mean and example-mean objectives, achieving **25% fewer training updates** than independent full training.
- Designed prospective validation-based selection experiments and verified deterministic checkpoint/resume continuation across model, optimizer, and RNG states.
- Evaluated the approach on **3 prospective random seeds**, observing agreement with full-training validation selection in 3/3 cases while documenting generalization limitations.

Tiny Verifiable LLM Reinforcement Learning Environment *Python, PyTorch, RL, Transformers*

- Built and executed a **19.9K-parameter decoder-only Transformer** for action prediction in a deterministic 6×6 Sokoban environment.
- Implemented a complete **BFS → supervised behavior cloning → on-policy REINFORCE** pipeline with executable environment interaction and verifier-based progress rewards.
- Conducted controlled experiments comparing **uniform and curriculum-based task sampling** using held-out solve rate, training behavior, and difficulty progression.
- Added automated testing, experiment configuration, reproducible training runs, and result reporting for CPU-based local execution.

LLM Inference Optimization Lab *Python, LLM Systems, KV Cache*

- Built deterministic simulators for LLM serving mechanisms including **request scheduling, KV-cache allocation, and prefill/decode batching**.
- Implemented baseline-versus-candidate policy evaluation with correctness validation, bounded configuration search, held-out workloads, and reproducible experiment receipts.
- Analyzed latency, throughput, cache growth, and batching trade-offs using abstract simulation metrics while separating simulation evidence from real GPU benchmarks.

Validation-Guided LoRA Loss Selection *Qwen3-0.6B, PyTorch, LoRA*

- Reproduced a controlled study comparing **token-mean vs. example-mean training objectives** for parameter-efficient fine-tuning of Qwen3-0.6B on GSM8K.
- Implemented a validation-guided objective-selection protocol with held-out test evaluation and safeguards against test-set leakage.
- Conducted paired experiments across multiple random seeds, comparing token-weighted NLL, example-level NLL, and exact-answer accuracy.
- Analyzed cases where the validation-selected objective failed to transfer to the test set, highlighting limitations of validation-based objective selection in constrained low-data settings.

Transformer From Scratch *PyTorch*

- Implemented the Transformer architecture from first principles in PyTorch, including token embeddings, positional encod-

ing, Query-Key-Value representations, scaled dot-product self-attention, and multi-head attention.
- Studied the relationship between attention mechanisms, contextual representations, and model architecture as preparation for LLM research.

Research & Technical Activities

- Developing reproducible experimental pipelines for LLM fine-tuning, reinforcement learning, hallucination detection, and inference optimization, with emphasis on controlled evaluation and multi-seed experiments.
- Implementing and evaluating Transformer-based models, LoRA/QLoRA fine-tuning methods, KV-cache mechanisms, and reinforcement-learning environments to investigate model behavior and training efficiency.
- Maintaining research code and experiment configurations with automated testing, reproducible runs, checkpoint validation, and held-out evaluation.

Professional Experience

SupMentr Technologies Pvt. Ltd., Bengaluru Feb 2026 – May 2026
Full Stack Web Development Intern – MERN

- Developed ChatApp, a full-stack real-time messaging application using MongoDB, Express.js, React, Node.js, and Socket.IO.
- Implemented JWT authentication, online/offline presence, message delivery and seen status, instant notifications, and user profiles.
- Designed REST APIs and MongoDB/Mongoose data models and integrated client-server communication using WebSockets.
- Developed the frontend using React, TypeScript, Vite, and Tailwind CSS.
- Integrated Cloudinary for image sharing and Multer for file handling.

Technical Skills

Programming: Python, Java, C, JavaScript, TypeScript
Machine Learning & AI: PyTorch, Scikit-learn, TensorFlow.js, Hugging Face Transformers, Natural Language Processing, Large Language Models, Reinforcement Learning
LLM & Generative AI: Transformers, Self-Attention, KV Cache, LoRA, QLoRA, PEFT, 4-bit Quantization, LLM Fine-Tuning, RAG, Embeddings, Hallucination Detection, Distribution Shift
Research & Evaluation: BERT, BERTScore, Biomedical NER, NLL, Exact-Match Accuracy, Statistical Analysis, Experimental Design, Multi-Seed Evaluation, Held-Out Evaluation, Reproducible Experiments
Data & Databases: NumPy, Pandas, SQL, MySQL, SQLite, MongoDB, PostgreSQL, ChromaDB
Software & Tools: React.js, Node.js, Express.js, Next.js, REST APIs, Tailwind CSS, Git, GitHub, GitLab, Docker, AWS, Jupyter Notebook, VS Code, Postman

Achievements

- **Top 6 Finalist** – CTRL+MOVE 24-Hour Hackathon (2025), selected among 105 participating teams.
- Participated in Code Carnage 24-Hour Hackathon (2025).
- Participated in AI Hiring Show: Vibe Coding, Power Hiring (2025).

Technical Writing

Published technical articles on Hashnode covering DNS resolution, Git internals, TCP handshakes, and browser internals.
hashnode.com/@bibekbhandari1

Additional Training

- Node.js – KnowledgeGate.in
- The Complete JavaScript Course 2025 – Udemy
- Introduction to Databases and SQL Querying – Udemy

Teaching Experience

Mathematics & Science Instructor *New Star English Boarding School, Sukhad, Kailali, Nepal | Jun 2026 – Sept 2026*

- Taught Mathematics and Science to Grades 9 and 10, delivering lessons and guiding students through conceptual understanding, problem-solving, and examination preparation.
- Designed practice problems, classroom activities, and assessments to reinforce core concepts and analytical problem-solving skills.
- Adapted explanations and instructional approaches to accommodate different levels of student understanding and learning needs.
- Provided individual guidance and feedback to help students identify conceptual gaps and improve problem-solving performance.